

EDUCATION

Purdue University, USA

Master of Science, Mechanical Engineering (minor: Computational Science)

GPA: 3.67/4 | Aug 2014 – Dec 2016

Vellore Institute of Technology, India

Bachelor of Technology, Mechanical Engineering

GPA: 9.03/10 | Aug 2010 – Jun 2014

SKILLS

Thermal Engineering

- Thermal validation & testing
- Touch-temperature limits & user perception studies
- Virtual temperature sensors (skin-hotspot prediction)
- Thermal solutions: graphite spreaders, foams, TIMs
- IR imaging, thermocouples & DAQ
- Software thermal management policies

Simulation & CAD

- Ansys Icepak
- FloTherm XT
- COMSOL
- SolidWorks CAD

Product Development & Programming

- Consumer electronics development (xVT)
- Use-case & power budget testing
- Test automation frameworks
- Python & Unix scripting
- MATLAB & data analysis

PUBLICATIONS

Experimental investigation of Phase Change Materials for thermal management of handheld devices

International Journal of Thermal Sciences (2018)

Investigation of aluminum foams and graphite fillers for improving the thermal conductivity of paraffin wax-based phase change materials

IEEE ITherm (2017)

Passive Thermal Management Using Phase Change Materials: Experimental Evaluation of Thermal Resistances

ASME InterPACK (2015)

SUMMARY

Thermal engineer who ships consumer electronics. 8+ years taking products from first concept through mass production — thermal validation (IR imaging, thermocouples, automated test frameworks), thermal simulation, evaluation of thermal solutions, and software thermal management policies that balance performance with thermal comfort and safety.

PROFESSIONAL EXPERIENCE

Apple	Aug 2023 – Present
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Thermal Engineer (System Product Design)

Thermal testing and design of iPad Air (6th & 7th gen, 2024/2025) and iPhone 17 (2025).

- Own all thermal objectives from initial concept through launch, defining software thermal management policies that balance performance and thermal comfort
- Characterize thermal performance across real-world use cases — benchmarks, gaming, video streaming, charging, camera — and analyze field metrics to mirror customer usage in tests
- Define touch-temperature limits through user perception studies; design virtual temperature sensors that predict skin hotspots and cut test time
- Evaluate feasibility and performance of thermal solutions: graphite spreaders, graphite-wrapped foams, copper sheets and TIMs
- Partner with power teams to define and validate use-case power budgets, driving EE optimizations when power exceeds targets
- Commissioned a thermal validation lab at a vendor site and authored thermal test SOPs

Amazon Lab126	Dec 2018 – Jul 2023
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Thermal Engineer II

Thermal testing and design of Echo Dot (3rd & 5th gen), Echo (3rd gen), Echo Show 10, Echo Studio, Echo Flex, Halo Rise and the Astro home robot.

- Enabled \$250k cost savings for Echo Dot by validating and assessing thermal risk during new product development
- Led thermal validation of Amazon's first consumer robot (Astro) against TS-1/TS-2 thermal safety specs
- Developed virtual temperature sensors to predict skin hotspots across safety validation tests, reducing test time
- Built a Python test-automation framework that cut test time by 50% by automating music/video streaming, display and LED control, and internal temperature and power capture
- Developed the Temperature Sensor feature on Echo Dot (5th gen) by collecting and processing large-scale device data with Python and Unix scripting

Syscon International	Feb 2017 – Oct 2018
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Product Engineer

Developed portable temperature measurement systems from concept to manufacturing for the Iron & Steel industry.

- Standardized thermocouple calibration; tested and troubleshoot hardware and software prototypes
- Sped up product builds by 25% with MATLAB algorithms for processing field data
- Trained distributors and sales teams; presented at AFS and AIST industry conferences

Purdue University	Jan 2015 – Dec 2016
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Graduate Research Assistant

Master's thesis: cooling smartphones using Phase Change Materials.

- Developed and validated a passive cooling solution (COMSOL, Ansys Fluent, IR camera, thermocouples), cutting processor temperatures by 5%
- Published in the International Journal of Thermal Sciences; presented at ASME InterPACK